MILITARY SPECIFICATION

# RECTIFIER, SILICON, DIRECT CURRENT, TRAINER POWER SUPPLY, TYPE F-1

## 1. SCOPE

* 1.1 This specification covers one type of portable, 50-ampere, 28-volt, direct current (DC), trainer power supply silicon rectifier, designated Type F-1.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | Box, Wood, Cleated-Plywood |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-C-3432 | Cable And Wire, Electrical (Power And Control, Flexible And Extra Flexible 300 And 600 Volts) |
| MIL-P-6457 | Power Supply, Metallic Rectifier Type, General Specification For |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements For Equipment |
| MIL-STD-1186 | Cushioning, Anchoring, Bracing, Blocking, And Waterproofing, With Appropriate Test Methods |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-P-6457 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

* 3.2.1 Exceptions to the requirements of MIL-P-6457. All references to metallic rectifiers shall be deleted and silicon rectifiers substituted. MIL-STD-461 shall apply for the electromagnetic interference tests.

3.3 Design and construction.

3.3.1 Configuration. The rectifier shall be designed and constructed as a portable unit with a configuration as shown on Figure 1.

3.3.1.1 Wheels. Two rubber-tired wheels with a minimum diameter of 8 inches shall be provided. The load on each wheel shall not exceed its rated load limit.

3.3.1.2 Handle. A handle shall be provided to enable personnel to maneuver the rectifier. The handle shall not interfere with the power cables or instruments.

3.3.1.3 Cable racks. Cable racks shall be provided for mounting the alternating- and direct-current power cables and shall be designed to permit complete portability of the rectifier.

* 3.3.1.4 Rectifiers. The rectifiers shall be silicon diodes and shall be connected for full wave rectification. When the rectifier is operated continuously in an ambient temperature of 55° Centigrade (C) at full rated load, the temperature of the diodes shall not exceed 120°C.

* 3.3.1.5 Input circuit breaker. A 115/220 volt alternating current (AC), time delay, magnetic-hydraulic circuit breaker shall be provided in the input circuit for overload and starting purposes. Without tripping, the circuit breaker shall meet the surge rating requirements of 4.6.8 of MIL-P-6457 and the intermittent duty requirements of 4.6.11 of MIL-P-6457. In addition, when an overload of 400 percent is applied, the circuit breaker shall trip before any damage is incurred by the unit.

* 3.3.1.6 Standard parts. AN or MIL standard parts shall be used where they suit the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, et cetera, may be used provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN or MIL part is in effect on the date of invitation for bids, commercial parts may be used, provided written approval is obtained from the cognizant engineering activity. Use of nonapproved, nonstandard components will cause rejection of first article.



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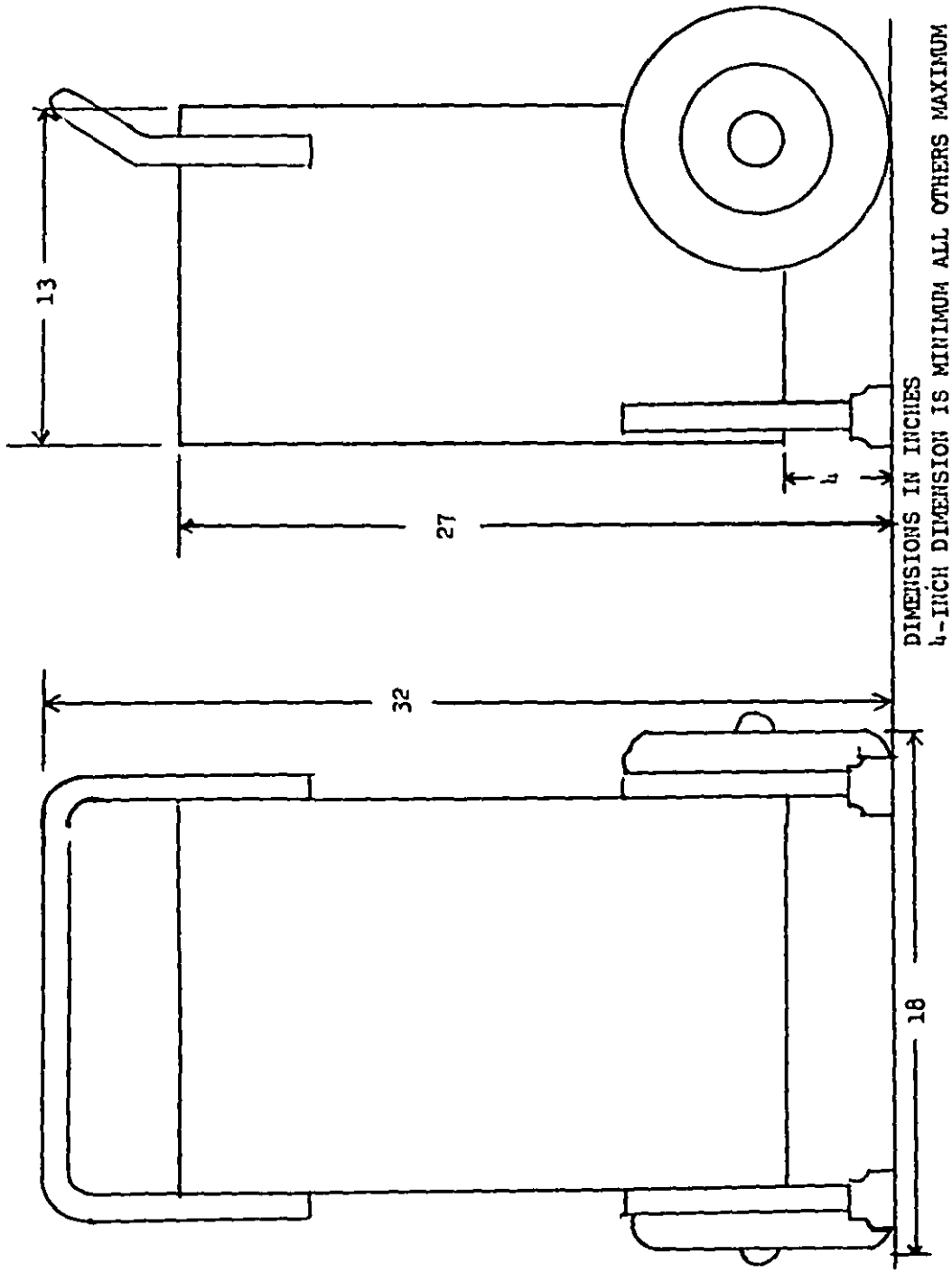



FIGURE 1. RECTIFIER CONFIGURATION



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3.3.2 Electrical characteristics.

3.3.2.1 Input voltage. The rectifier shall be designed to operate from an input voltage of 115 volts, 60 cycles, single-phase and shall be reconnectible to operate from 230 volts, 60 cycles, single-phase.

3.3.2.2 Output rating. The continuous direct-current output rating of the rectifier shall be 50 amperes at 28 volts.

3.3.2.3 Efficiency. The efficiency of the rectifier shall be not less than 70 percent.

3.3.2.4 Regulation. With the input voltage remaining constant at the nominal rated value, the no-load voltage at the output terminals shall remain within the voltage band of 28.0 to 30.5 volts. The output voltage shall vary not more than 2.5 volts as the load is changed from no-load to full load.

* 3.3.2.5 Ripple voltage. The peak alternating-current ripple voltage superimposed on the direct current output voltage shall not exceed 1.2 volts peak when determined for both polarities as specified in MIL-P-6457.

3.3.2.6 Feedback protection. The rectifier circuitry shall be designed to avoid feedback through bleeder resistors if the rectifier is connected in parallel with another direct-current power supply or battery system.

3.4 Performance.

3.4.1 Operating temperatures. The rectifier shall be capable of operating under all temperature conditions from -40° to +50°C.

3.5 Power cables.

3.5.1 Input cable. The rectifier shall be provided with a permanently attached, 3-wire, alternating-current power cable. The cable shall be type I as specified in MIL-P-6457. It shall be 50 feet long and shall have a maximum line voltage drop of 1 percent rated voltage at continuous rated load (for the 115 volt connection).

3.5.2 Output cable. The rectifier shall be provided with one each 2-conductor, No. 4 AWG, class HO, flexibility E banded cable in accordance with MIL-C-3432. The cable shall be 186 ±2 inches long. The cable sheath on each conductor shall be stripped back a distance of 5 ±1 inch to facilitate attachment of a connector. The black wire shall be connected to the positive terminal of the rectifier. The white wire shall be connected to the negative terminal of the rectifier.

3.6 Meters. Commercial-type meters with a 2 percent accuracy shall be provided. The scale range of the direct-current ammeter shall be 0 to 75 amperes, and the scale range of the direct-current output voltmeter shall be 0 to 40 volts DC.

3.7 Dimensions. The dimensions of the rectifier shall be in accordance with Figure 1.

3.8 Weight. The total net weight of the rectifier, exclusive of external cables, shall not exceed 100 pounds.



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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Additional preproduction tests. The tests listed under 4.2 are preproduction tests and shall be conducted in addition to the preproduction tests of MIL-P-6457 as specified in 4.1.

* 4.2 Test methods.

* 4.2.1 Silicon rectifier temperature rise. The rectifier shall be operated at rated load in an ambient temperature of 55°C until thermal equilibrium is reached. Thermocouples shall be placed in at least three places in the silicon diodes to determine that the temperature of the diodes does not exceed 120°C. If desired, this test may be combined with the high temperature test specified in 4.6.15a of MIL-P-6457.

* 4.2.2 Input circuit breaker test. The rectifier shall be operated in an ambient temperature of 25°C with an input of 115 volts AC and then reconnected for 220 volt AC operation. It shall be shown that with each input voltage, when a 400 percent load is applied, that the input circuit breaker will trip before any damage is incurred by the unit. Immediately at the conclusion of this test, the rectifier shall meet the regulation and efficiency requirements as specified in 4.6.4 and 4.6.5 of MIL-P-6457. If desired, this test may be combined with the operational test specified in 4.6.3 of MIL-P-6457.

* 4.3 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Rectifiers shall be preserved and packaged in accordance with MIL-P-116, method IIA, one each per unit container. The unit container shall be as specified in 5.2.1 and 5.2.2 below.

5.1.2 Cleaning and drying. Cleaning and drying shall be in accordance with applicable processes and procedures of MIL-P-116.

5.1.3 Level C. Rectifiers shall be afforded cleaning, drying, preservation, and packaging which affords adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.



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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Rectifiers preserved and packaged to meet 5.1.1 shall be packed in overseas-type shipping containers conforming to PPP-B-601. As far as practicable, shipping containers shall be of uniform shape and size, and of mininum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the specification and appendix thereto.

5.2.2 Level B. Rectifiers preserved and packaged to meet 5.1.1 shall be packed in domestic-type shipping containers conforming to PPP-B-601. As far as practicable, shipping containers shall be of uniform shape and size, and of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the specification and appendix thereto.

5.2.3 Level C. Packing shall be of the type which affords protection at the lowest rate against damage during direct domestic shipments from supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type F-1 rectifier is intended for use as a source of direct-current electrical power for trainers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

c. Special shipment marking (see 5.4).

* 6.3 The outside margins of this specification have been marked to indicate where changes (deletions, additions, et cetera) from the previous issue have been made. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Preparing Activity:
Air Force - 82

Review Activity:
Air Force - 82

Project No. 6130-F092



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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